

Section 5: Solids, liquids and gases

- a) Units
- b) Density and pressure
- c) Change of state
- d) Ideal gas molecules

a) Units

Students will be assessed on their ability to:

- 5.1 use the following units: degrees Celsius ($^{\circ}\text{C}$), kelvin (K), joule (J), kilogram (kg), kilogram/metre³ (kg/m³), metre (m), metre² (m²), metre³ (m³), metre/second (m/s), metre/second² (m/s²), newton (N), pascal (Pa).

b) Density and pressure

Students will be assessed on their ability to:

- 5.2 know and use the relationship between density, mass and volume:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$\rho = \frac{m}{V}$$

- 5.3 describe experiments to determine density using direct measurements of mass and volume
- 5.4 know and use the relationship between pressure, force and area:

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$p = \frac{F}{A}$$

- 5.5 understand that the pressure at a point in a gas or liquid which is at rest acts equally in all directions
- 5.6 know and use the relationship for pressure difference:

$$\text{pressure difference} = \text{height} \times \text{density} \times g$$

$$p = h \times \rho \times g$$

c) Change of state

Students will be assessed on their ability to:

- 5.7 understand the changes that occur when a solid melts to form a liquid, and when a liquid evaporates or boils to form a gas**
- 5.8 describe the arrangement and motion of particles in solids, liquids and gases**

d) Ideal gas molecules

Students will be assessed on their ability to:

- 5.9 understand the significance of Brownian motion, as supporting evidence for particle theory
- 5.10 understand that molecules in a gas have a random motion and that they exert a force and hence a pressure on the walls of the container
- 5.11 understand why there is an absolute zero of temperature which is -273°C
- 5.12 describe the Kelvin scale of temperature and be able to convert between the Kelvin and Celsius scales
- 5.13 understand that an increase in temperature results in an increase in the average speed of gas molecules
- 5.14 understand that the Kelvin temperature of the gas is proportional to the average kinetic energy of its molecules**
- 5.15 describe the qualitative relationship between pressure and Kelvin temperature for a gas in a sealed container
- 5.16 use the relationship between the pressure and Kelvin temperature of a fixed mass of gas at constant volume:**

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

- 5.17 use the relationship between the pressure and volume of a fixed mass of gas at constant temperature:

$$p_1V_1 = p_2V_2$$